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Role : SOC Analyst

Day 03 Task: Network Security Monitoring and Analysis

Objective:

Analysis by using Wireshark for passive traffic analysis, Snort for active intrusion detection, and Scapy for generating test traffic. You will also learn to identify and respond to common network security threats.

Topics to Study

- **Network Traffic Analysis:** Using Wireshark to capture, filter, and analyze network packets.
- **Intrusion Detection Systems (IDS):** Setting up and configuring Snort for basic IDS functionality.
- **Packet Generation:** Using Scapy to create specific network packets for testing IDS rules.
- **Common Network Attacks:** Understanding basic attacks like port scans, SYN floods, and reconnaissance.

TASK-1

Step 1: Download Wireshark

- Visit the official Wireshark website: <https://www.wireshark.org/>
- Choose the appropriate installer for your operating system:
 - **Windows:** .exe installer
 - **Linux:** Use the package manager (e.g., apt, dnf)
 - **macOS:** .dmg file
- Download and run the installer.
 - On Windows, during installation, you may be prompted to install **WinPcap** or **Npcap** — accept it. This is necessary for packet capturing.

Step 2: Install Wireshark

On Windows:

- Double-click the downloaded .exe file.
- Follow the setup wizard:
 - Select components to install (leave defaults)
 - Accept license agreements

- Allow USBPcap and Npcap installation

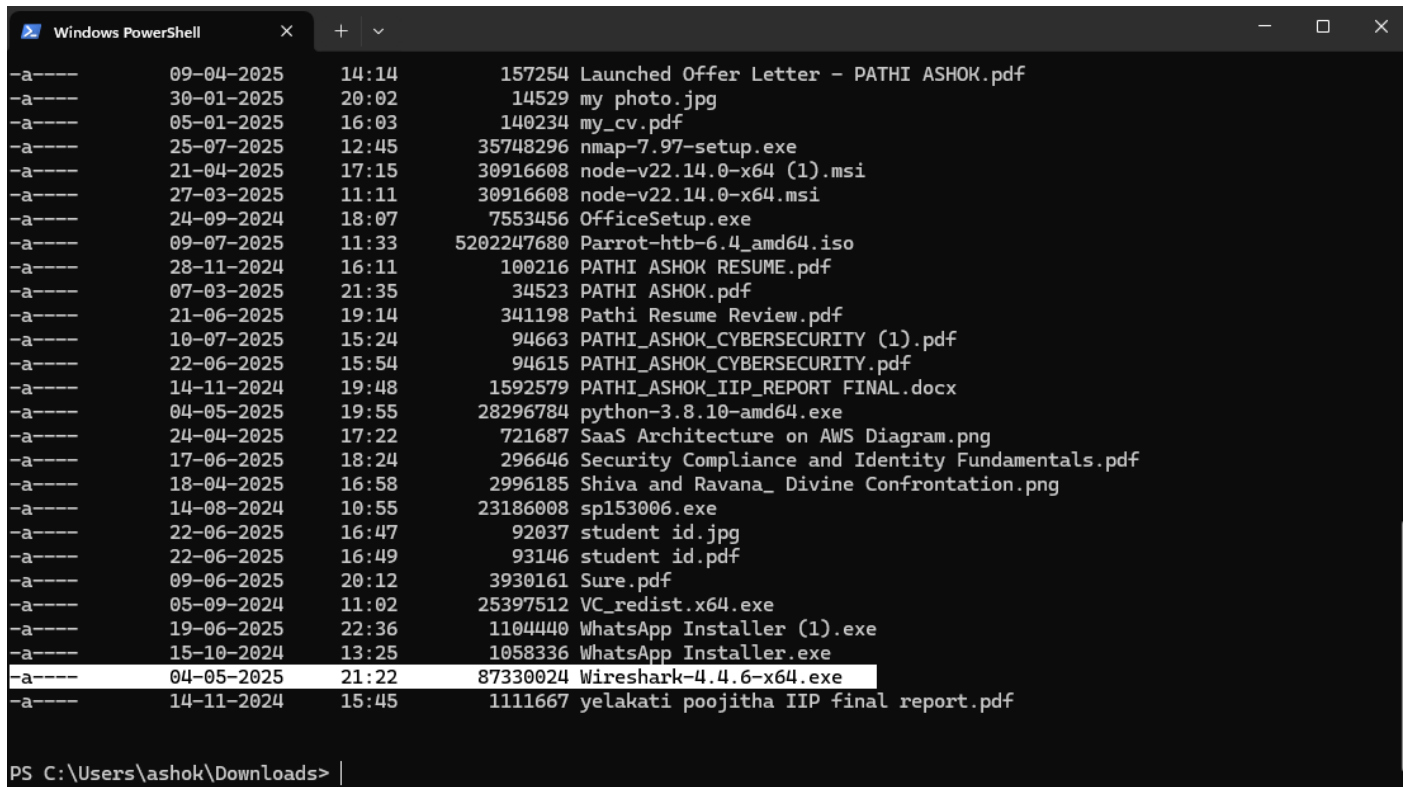
Commands To install wireshark:

1. `sudo apt update`
2. `sudo apt install wireshark -y`

Step 3: Verify Installation

- Open **Wireshark** from Start Menu (Windows) or Applications (Linux).
- Check that network interfaces are listed (e.g., Wi-Fi, eth0, lo).
- Begin a live capture to ensure functionality.

Open the terminal and type: `tshark -v`



```
Windows PowerShell
-a----- 09-04-2025 14:14 157254 Launched Offer Letter - PATHI ASHOK.pdf
-a----- 30-01-2025 20:02 14529 my photo.jpg
-a----- 05-01-2025 16:03 140234 my_cv.pdf
-a----- 25-07-2025 12:45 35748296 nmap-7.97-setup.exe
-a----- 21-04-2025 17:15 30916608 node-v22.14.0-x64 (1).msi
-a----- 27-03-2025 11:11 30916608 node-v22.14.0-x64.msi
-a----- 24-09-2024 18:07 7553456 OfficeSetup.exe
-a----- 09-07-2025 11:33 5202247680 Parrot-htb-6.4_amd64.iso
-a----- 28-11-2024 16:11 100216 PATHI ASHOK RESUME.pdf
-a----- 07-03-2025 21:35 34523 PATHI ASHOK.pdf
-a----- 21-06-2025 19:14 341198 Pathi Resume Review.pdf
-a----- 10-07-2025 15:24 94663 PATHI_ASHOK_CYBERSECURITY (1).pdf
-a----- 22-06-2025 15:54 94615 PATHI_ASHOK_CYBERSECURITY.pdf
-a----- 14-11-2024 19:48 1592579 PATHI_ASHOK_IIP_REPORT FINAL.docx
-a----- 04-05-2025 19:55 28296784 python-3.8.10-amd64.exe
-a----- 24-04-2025 17:22 721687 SaaS Architecture on AWS Diagram.png
-a----- 17-06-2025 18:24 296646 Security Compliance and Identity Fundamentals.pdf
-a----- 18-04-2025 16:58 2996185 Shiva and Ravana_ Divine Confrontation.png
-a----- 14-08-2024 10:55 23186008 sp153006.exe
-a----- 22-06-2025 16:47 92037 student id.jpg
-a----- 22-06-2025 16:49 93146 student id.pdf
-a----- 09-06-2025 20:12 3930161 Sure.pdf
-a----- 05-09-2024 11:02 25397512 VC_redist.x64.exe
-a----- 19-06-2025 22:36 1104440 WhatsApp Installer (1).exe
-a----- 15-10-2024 13:25 1058336 WhatsApp Installer.exe
-a----- 04-05-2025 21:22 87330024 Wireshark-4.4.6-x64.exe
-a----- 14-11-2024 15:45 1111667 yelakati poojitha IIP final report.pdf

PS C:\Users\ashok\Downloads> |
```

Fig 1 wireshark has been pre installed in my local virtual machine

Common filtered used in wire shark:

Filter	Description
ip	Show only IP packets
tcp	Show only TCP packets
udp	Show only UDP packets
icmp	Show only ping packets
http	Show only HTTP packets
tcp.port == 80	Show packets on TCP port 80

Filter

Description

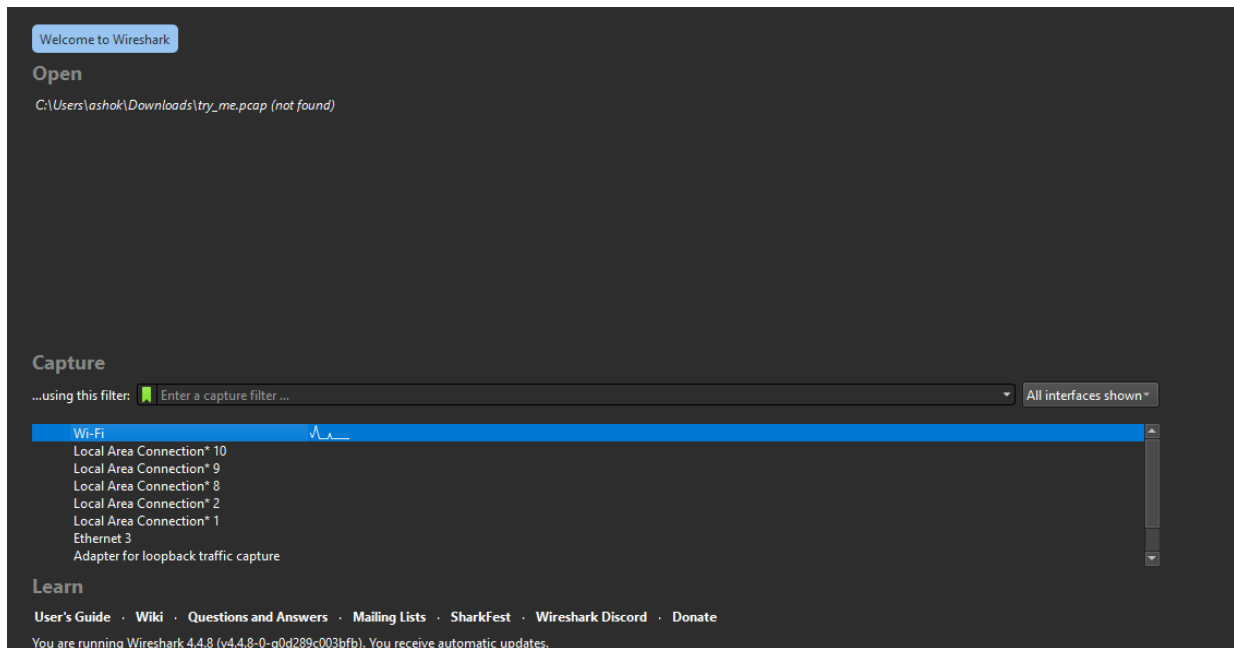
ip.addr == 192.168.1.1

Show traffic to/from specific IP

TASK – 2

Step 1: Start Packet Capture

- Launch Wireshark.



- Select a network interface such as:
 - Wi-Fi for general internet traffic
 - Loopback (lo) for local testing
- Click the interface to start capturing packets in real-time.
- Let the capture run for 10 minutes to gather sufficient data.

Step 2: Apply Protocol Filters

- Use the **filter bar** in Wireshark to filter by protocol:
 - tcp — for transport control packets (e.g., HTTP, SSH)

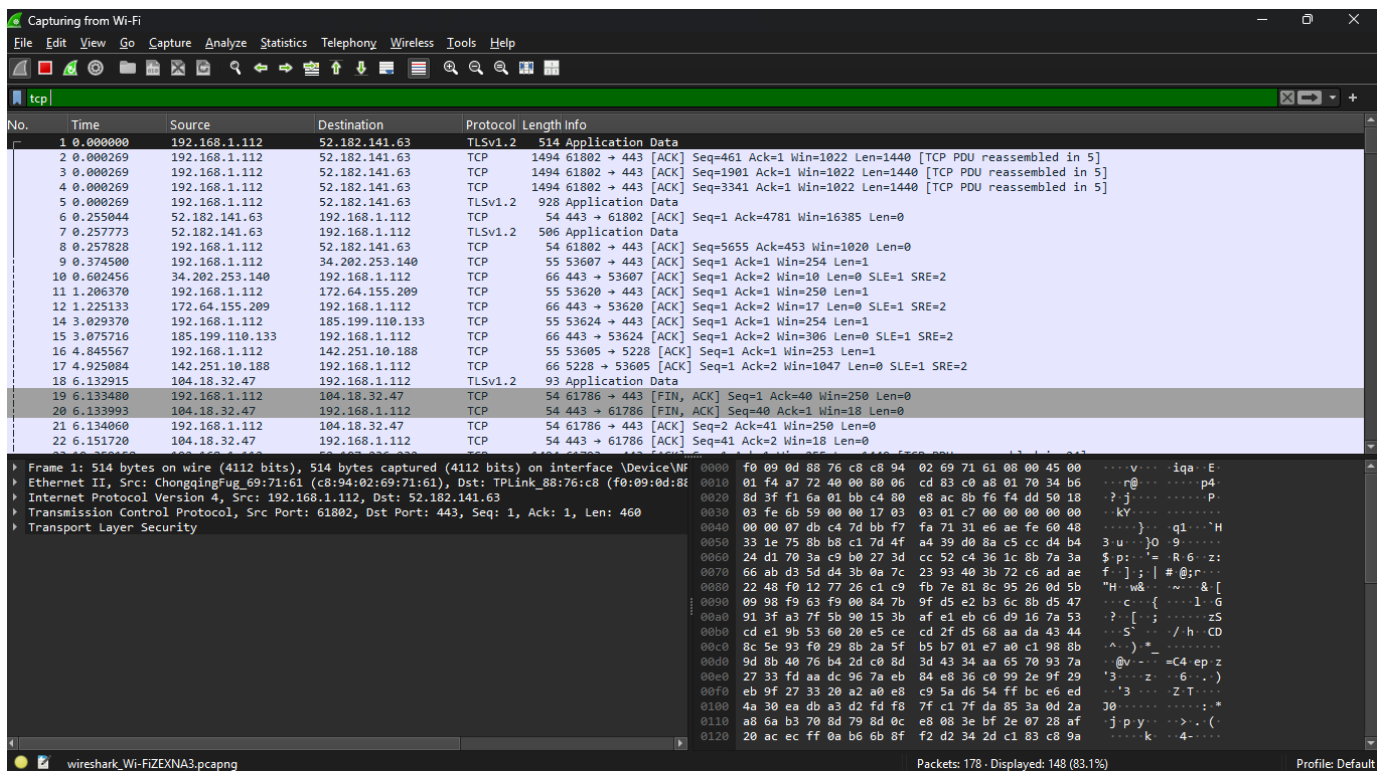


Fig 3: Tcp Filtered Packets

- o udp — for connectionless traffic (e.g., DNS, DHCP, SSDP)

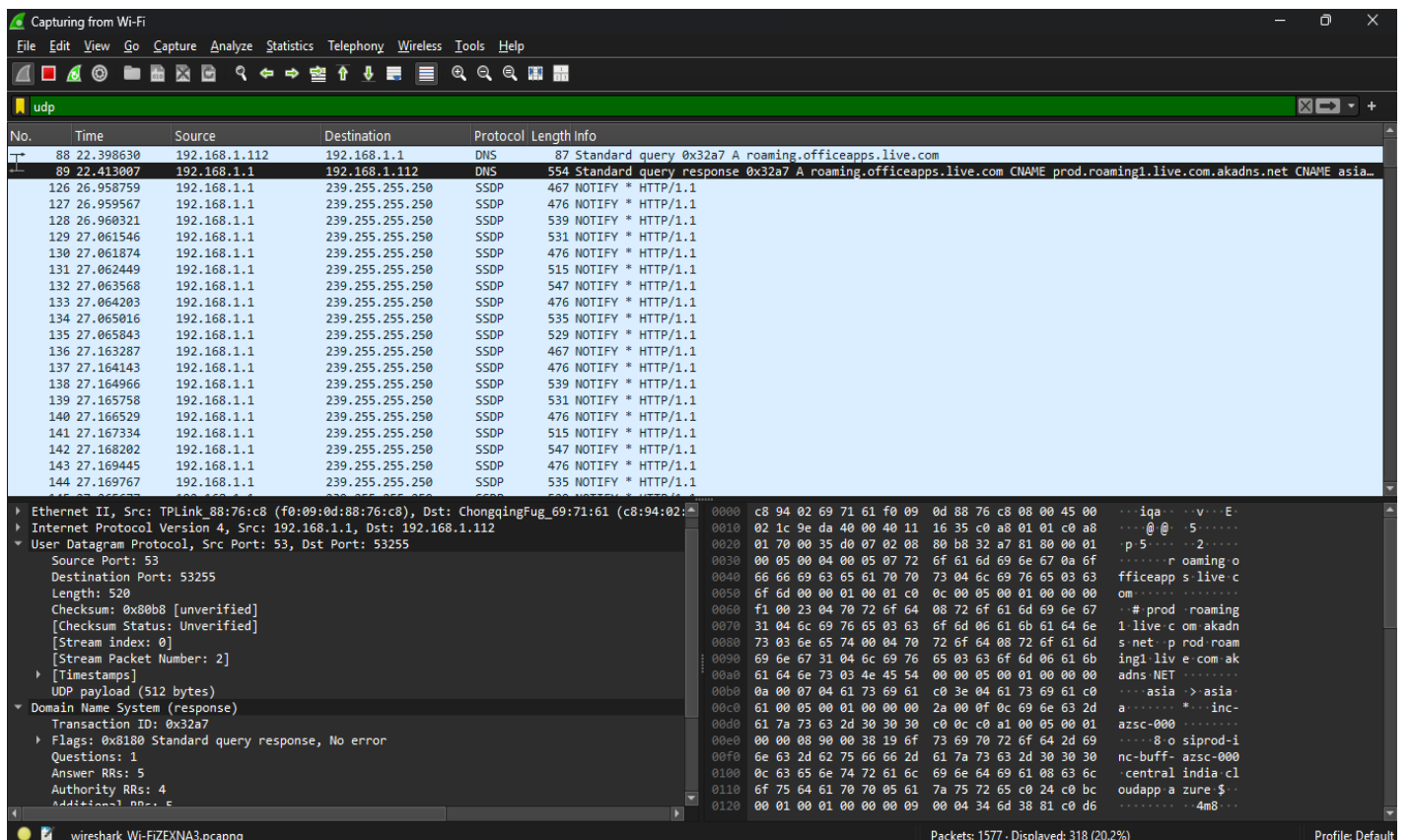


Fig 4 : Displayed UDP protocols under DNS and SSDP. Total display 318/ 1577 packets

- o icmp — for diagnostics like ping/traceroute

(THERE IS NO ICMP)

Step 3: Identify Protocols and Their Purposes

1. **TCP (Transmission Control Protocol)** — Ensures reliable, ordered delivery of data (used in HTTP, HTTPS, FTP).
2. **UDP (User Datagram Protocol)** — Lightweight protocol for fast communication (used in DNS, VoIP).
3. **ICMP (Internet Control Message Protocol)** — Used for network diagnostics (e.g., ping, traceroute).

Step 4: Look for Suspicious Traffic

Command : tcp.flags.syn == 1 && tcp.flags.ack == 0

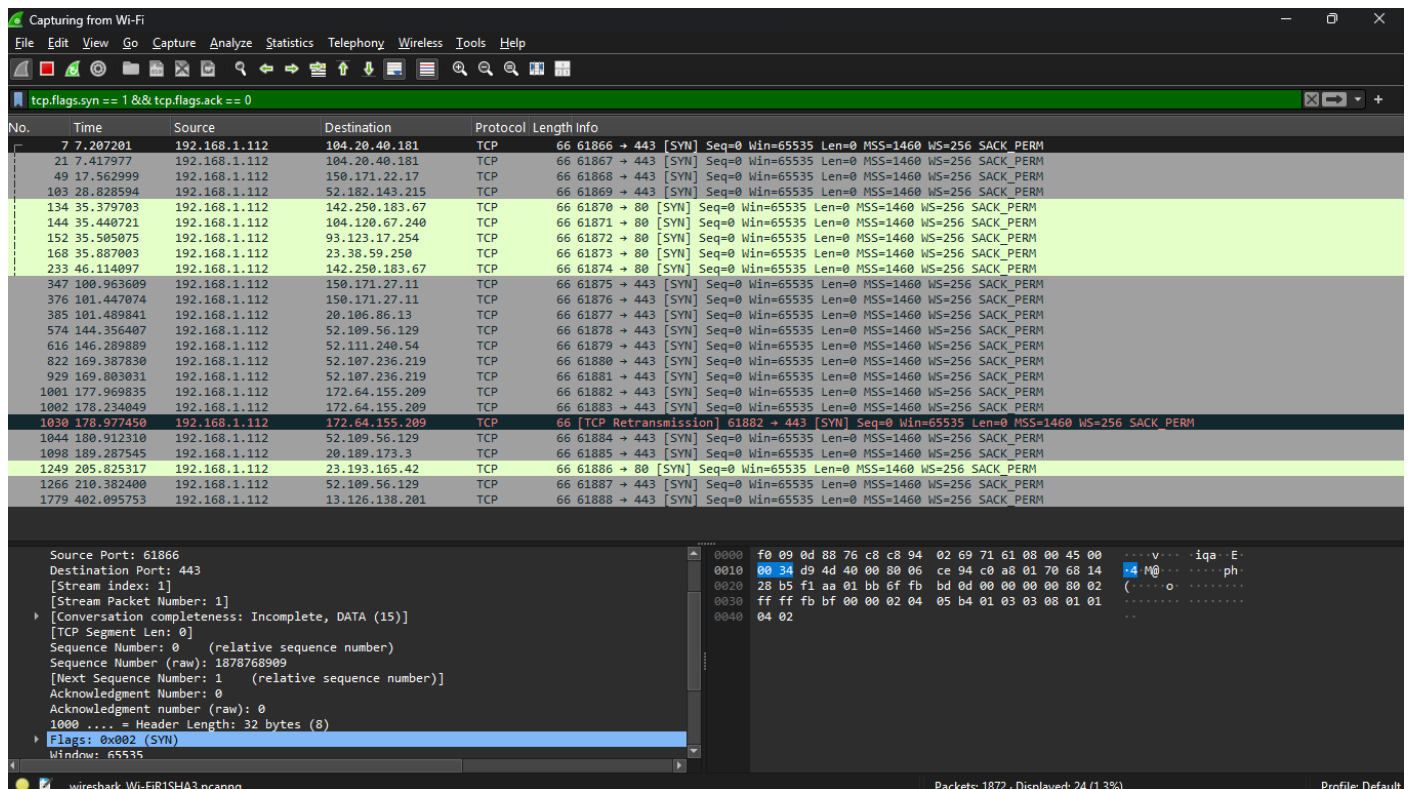


Fig 5 : tcp syn suspicious Traffic

Suspicious Patterns Observed:

- Multiple **SYN packets** from the same source IP targeting different ports — possible **port scan**.
- Sudden spikes in **ICMP echo requests** within a short time window — possibly a **ping flood** attempt.
- DNS request bursts to uncommon domains — potential sign of **DNS tunneling**.

TASK – 3

(Installing and Configuring Snort in IDS Mode)

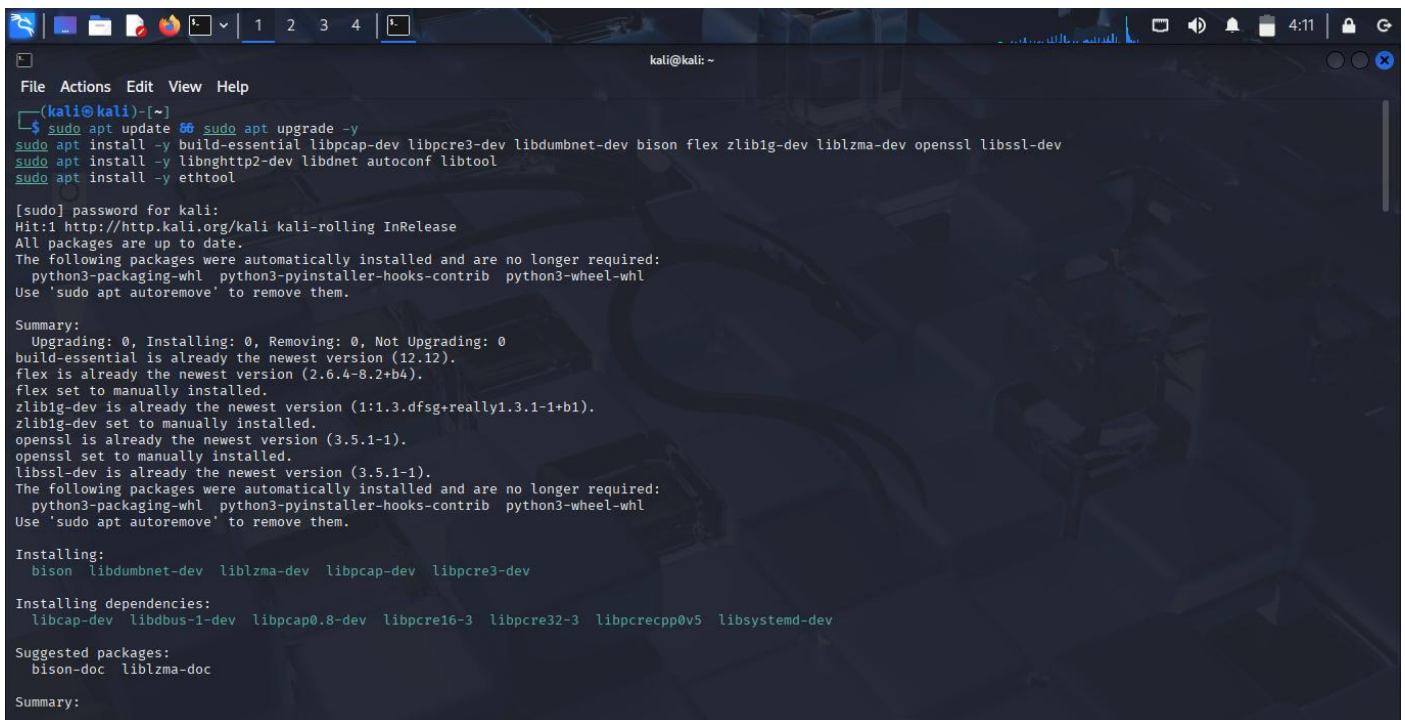
Step 1: Set Up a Linux VM

- Use VirtualBox, VMware, or WSL (Windows Subsystem for Linux).
- Install a clean version of Kali Linux.

Step 2: Install Dependencies

Commands :

- `sudo apt update && sudo apt upgrade -y`
- `sudo apt install -y build-essential libpcap-dev libpcr3-dev libdumbnet-dev bison flex zlib1g-dev liblzma-dev openssl libssl-dev`
- `sudo apt install -y libnghttp2-dev libdnet autoconf libtool`
- `sudo apt install -y ethtool`



```
(kali@kali)~$ sudo apt update && sudo apt upgrade -y
[sudo] password for kali:
Hit:1 http://http.kali.org/kali kali-rolling InRelease
All packages are up to date.
The following packages were automatically installed and are no longer required:
  python3-packaging-whl python3-pyinstaller-hooks-contrib python3-wheel-whl
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
build-essential is already the newest version (12.12).
flex is already the newest version (2.6.4-8.2+b4).
flex set to manually installed.
zlib1g-dev is already the newest version (1:1.3.dfsg+really1.3.1-1+b1).
zlib1g-dev set to manually installed.
openssl is already the newest version (3.5.1-1).
openssl set to manually installed.
libssl-dev is already the newest version (3.5.1-1).
The following packages were automatically installed and are no longer required:
  python3-packaging-whl python3-pyinstaller-hooks-contrib python3-wheel-whl
Use 'sudo apt autoremove' to remove them.

Installing:
  bison libdumbnet-dev liblzma-dev libpcap-dev libpcr3-dev

Installing dependencies:
  libcap-dev libdbus-1-dev libcap0.8-dev libpcr16-3 libpcr32-3 libpcrrecpp0v5 libsystemd-dev

Suggested packages:
  bison-doc liblzma-doc

Summary:
```

Fig 6 updating the kali linux and installing libssl

Step 3: Download and Install Snort

Command : `sudo apt install snort -y`


```
kali@kali: ~  
File Actions Edit View Help  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0  
  
(kali@kali)-[~]  
$ sudo apt install snort -y  
The following packages were automatically installed and are no longer required:  
  python3-packaging-whl python3-pyinstaller-hooks-contrib python3-wheel-whl  
Use 'sudo apt autoremove' to remove them.  
  
Installing:  
  snort  
  
Installing dependencies:  
  libdaq3 libestr0 libfastjson4 liblognorm5 oinkmaster rsyslog snort-common snort-common-libraries snort-rules-default  
  
Suggested packages:  
  rsyslog-doc | rsyslog-pgsql rsyslog-elasticsearch rsyslog-hiredis rsyslog-kubernetes rsyslog-clickhouse | rsyslog-gnutls rsyslog-relp  
  rsyslog-mysql rsyslog-mongodb rsyslog-kafka rsyslog-snmp rsyslog-docker rsyslog-openssl rsyslog-gssapi snort-doc  
  
Summary:  
  Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 0  
  Download size: 3,680 kB  
  Space needed: 15.8 MB / 40.8 GB available  
  
Get:1 http://http.kali.org/kali kali-rolling/main amd64 snort-common-libraries amd64 3.1.82.0-0kali1+b1 [269 kB]  
Get:2 http://mirror.aktkn.sg/kali kali-rolling/main amd64 snort-rules-default all 3.1.82.0-0kali1 [220 kB]  
Get:3 http://kali.download/kali kali-rolling/main amd64 snort-common all 3.1.82.0-0kali1 [117 kB]  
Get:4 http://kali.download/kali kali-rolling/main amd64 libestr0 amd64 0.1.11-2 [9,048 B]  
Get:5 http://kali.download/kali kali-rolling/main amd64 libfastjson4 amd64 1.2304.0-2 [28.9 kB]  
Get:6 http://kali.download/kali kali-rolling/main amd64 liblognorm5 amd64 2.0.6-5 [66.5 kB]  
Get:7 http://kali.download/kali kali-rolling/main amd64 rsyslog amd64 8.2504.0-1 [758 kB]  
Get:8 http://http.kali.org/kali kali-rolling/main amd64 libdaq3 amd64 3.0.12-0kali3+b1 [36.3 kB]  
Get:9 http://http.kali.org/kali kali-rolling/main amd64 snort amd64 3.1.82.0-0kali1+b1 [2,094 kB]  
Get:10 http://kali.download/kali kali-rolling/main amd64 oinkmaster all 2.0-4.2 [80.3 kB]  
Fetched 3,680 kB in 3s (1,380 kB/s)
```

Fig 7 installing the snort

Step 4: Verify Snort Installation

Command : snort -V

```
kali@kali: ~  
File Actions Edit View Help  
Setting up libestr0:amd64 (0.1.11-2) ...  
Setting up libfastjson4:amd64 (1.2304.0-2) ...  
Setting up snort-common (3.1.82.0-0kali1) ...  
Setting up liblognorm5:amd64 (2.0.6-5) ...  
Setting up libdaq3 (3.0.12-0kali3+b1) ...  
Setting up snort-rules-default (3.1.82.0-0kali1) ...  
Setting up snort-common-libraries (3.1.82.0-0kali1+b1) ...  
Setting up rsyslog (8.2504.0-1) ...  
Created symlink '/etc/systemd/system/syslog.service' → '/usr/lib/systemd/system/rsyslog.service'.  
Created symlink '/etc/systemd/system/multi-user.target.wants/rsyslog.service' → '/usr/lib/systemd/system/rsyslog.service'.  
Setting up snort (3.1.82.0-0kali1+b1) ...  
snort.service is a disabled or a static unit, not starting it.  
Processing triggers for man-db (2.13.1-1) ...  
Processing triggers for kali-menu (2025.3.0) ...  
Processing triggers for libc-bin (2.41-9) ...  
  
(kali@kali)-[~]  
$ snort -V  
  
o'--)-> Snort++ <*-  
....  
Version 3.1.82.0  
By Martin Roesch & The Snort Team  
http://snort.org/contact#team  
Copyright (C) 2014-2024 Cisco and/or its affiliates. All rights reserved.  
Copyright (C) 1998-2013 Sourcefire, Inc., et al.  
Using DAQ version 3.0.12  
Using LuaJIT version 2.1.1700206165  
Using OpenSSL 3.5.1 1 Jul 2025  
Using libpcap version 1.10.5 (with TPACKET_V3)  
Using PCRE version 8.39 2016-06-14  
Using ZLIB version 1.3.1  
Using LZMA version 5.8.1
```

Fig 8 verifying the snort installation

Commands :

- cd /etc/snort
- sudo nano /etc/snort/rules/local.rules

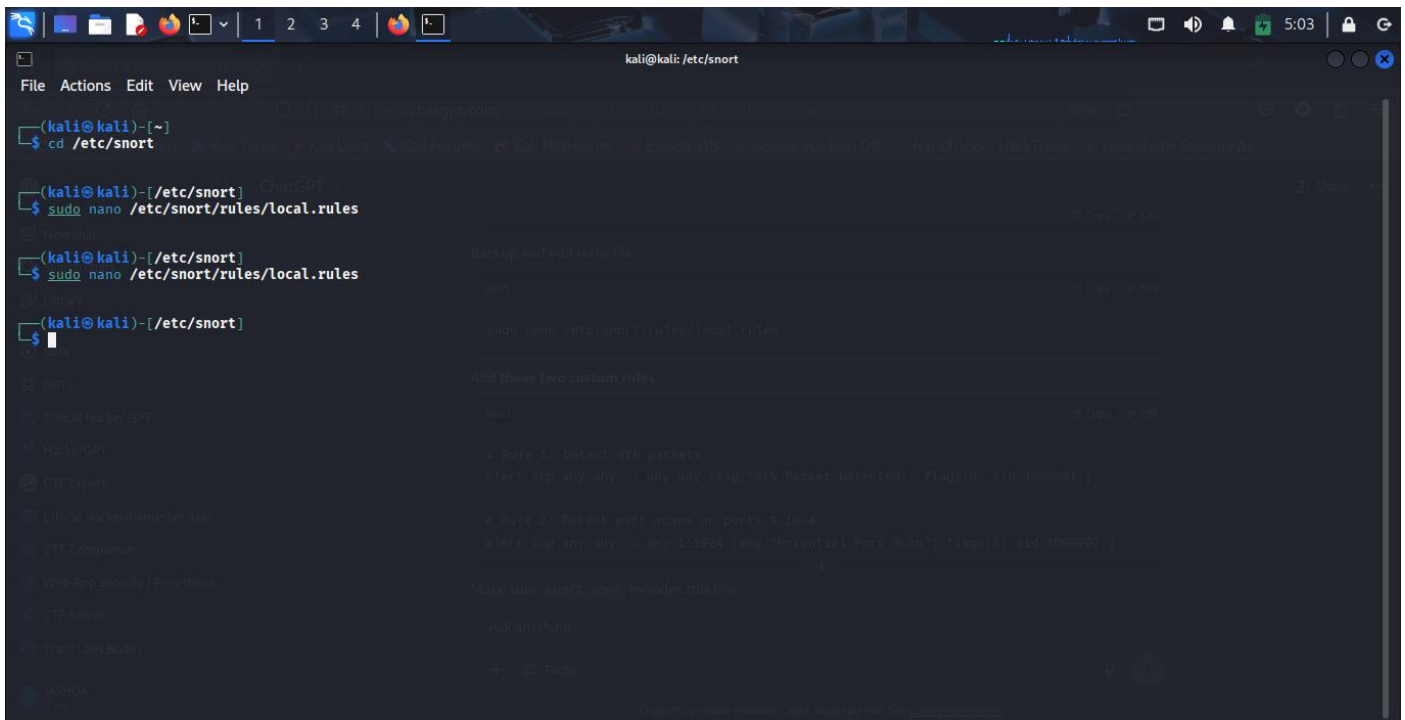


Fig 9 accessing the snort and local rules

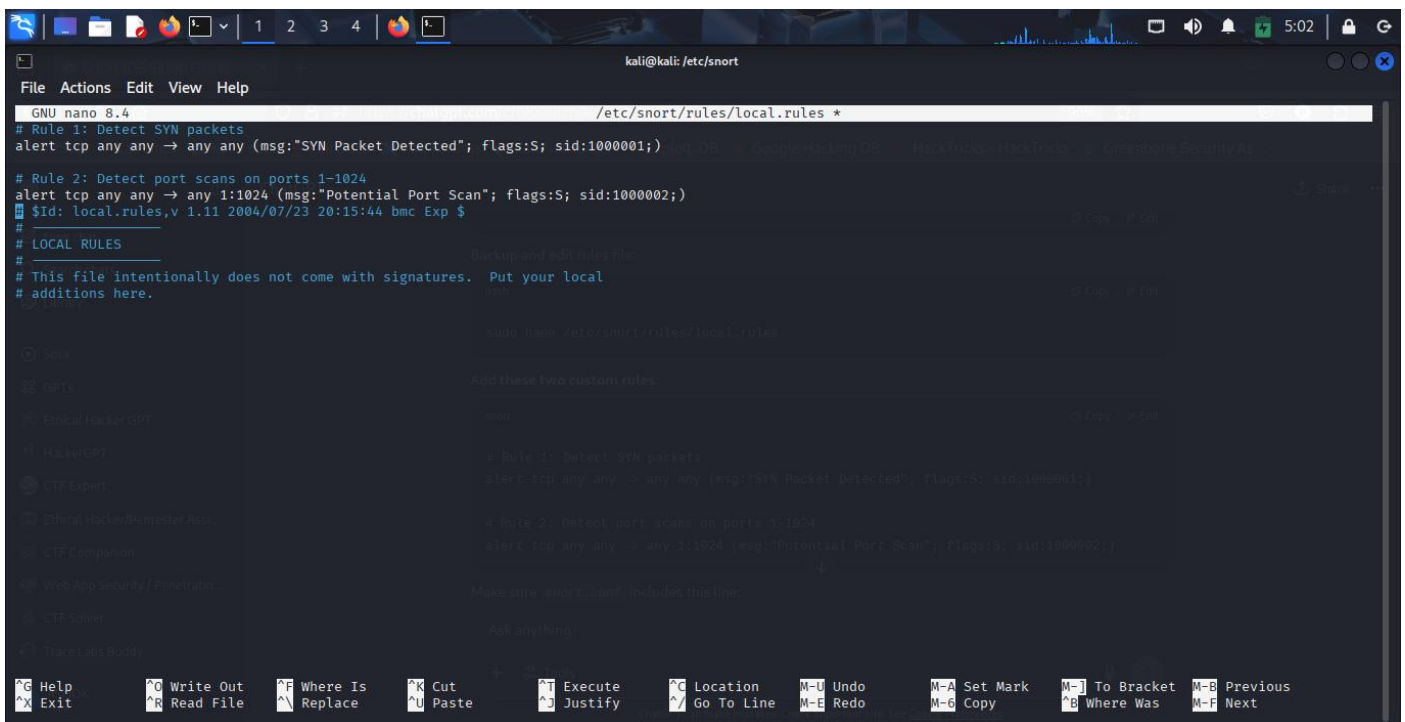


Fig 10 updating the rules of syn and port scan packets in local rules

Step 5: Configure Snort to Run in IDS Mode

alert tcp any any -> any any (msg:"SYN Packet Detected"; flags:S; sid:1000001;)

alert tcp any any -> any 1:1024 (msg:"Potential Port Scan"; flags:S; sid:1000002;)

Command : `sudo snort -i eth0 -c /etc/snort/snort.conf -A console`

`sudo` — Runs as root (required for packet capture)

`-i eth0` — Monitors the eth0 interface

-c /etc/snort/snort.conf — Uses Snort configuration and rules

-A console — Displays alerts directly in the terminal

Output:

07/29-12:14:25.123456 [**] [1:1000001:0] SYN Packet Detected [**]

07/29-12:14:25.123456 [Classification: Attempted Information Leak] [Priority: 2]

{TCP} 192.168.1.101:50830 -> 192.168.1.1:22

Task 4 : Generate Test Traffic with Scapy

File name : test_packets.py

```
from scapy.all import IP, TCP, send
from typing import List

def send_syn_scan_packets(target_ip: str, ports: List[int]) -> None:
    for port in ports:
        pkt = IP(dst=target_ip) / TCP(dport=port, flags="S")
        send(pkt, verbose=False)
        print(f"[+] Sent SYN to {target_ip}:{port}")

if __name__ == "__main__":
    target_ip = "192.168.1.10"
    target_ports = [21, 22, 23, 25, 80, 443, 8080]
    send_syn_scan_packets(target_ip, target_ports)
```

output:

WARNING: MAC address to reach destination not found. Using broadcast.

[+] Sent SYN to 192.168.1.10:21

WARNING: MAC address to reach destination not found. Using broadcast.

[+] Sent SYN to 192.168.1.10:22

WARNING: more MAC address to reach destination not found. Using broadcast.

[+] Sent SYN to 192.168.1.10:23

WARNING: MAC address to reach destination not found. Using broadcast.

[+] Sent SYN to 192.168.1.10:25

WARNING: MAC address to reach destination not found. Using broadcast.

[+] Sent SYN to 192.168.1.10:80

WARNING: more MAC address to reach destination not found. Using broadcast.

[+] Sent SYN to 192.168.1.10:443

WARNING: MAC address to reach destination not found. Using broadcast.

[+] Sent SYN to 192.168.1.10:8080

PS C:\Users\ashok\OneDrive\Desktop\CYART PROJECT\DAY 03>

```
5 def send_syn_scan_packets(target_ip: str, ports: List[int]) -> None:
6
7     for port in ports:
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
PS C:\Users\ashok\OneDrive\Desktop\CYART PROJECT\DAY 03>
PS C:\Users\ashok\OneDrive\Desktop\CYART PROJECT\DAY 03> c:: cd 'c:\Users\ashok\OneDrive\Desktop\CYART PROJECT\DAY 03'; & 'c:\Users\ashok\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Users\ashok\.vscode\extensions\ms-python.debugpy-2025.10.0-win32-x64\
bundled\libs\debugpy\launcher' '46038' '--' 'C:\Users\ashok\OneDrive\Desktop\CYART PROJECT\DAY 03\test_packets.py'
WARNING: MAC address to reach destination not found. Using broadcast.
[+] Sent SYN to 192.168.1.10:21
WARNING: MAC address to reach destination not found. Using broadcast.
[+] Sent SYN to 192.168.1.10:22
WARNING: more MAC address to reach destination not found. Using broadcast.
[+] Sent SYN to 192.168.1.10:23
WARNING: MAC address to reach destination not found. Using broadcast.
[+] Sent SYN to 192.168.1.10:25
WARNING: MAC address to reach destination not found. Using broadcast.
[+] Sent SYN to 192.168.1.10:80
WARNING: more MAC address to reach destination not found. Using broadcast.
[+] Sent SYN to 192.168.1.10:443
WARNING: MAC address to reach destination not found. Using broadcast.
[+] Sent SYN to 192.168.1.10:8080
PS C:\Users\ashok\OneDrive\Desktop\CYART PROJECT\DAY 03>
```

+ ... [] x
powershell
Python Deb...

Ln 14, Col 53 Spaces: 4 UTF-8 CRLF Python Finish Setup Python 3.11